EE336 Fundamental of Photovoltaics

Lab I

Sun Position Calculation

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2026/03/06

Quiz1

A solar panel is installed parallel to the ground in Shenzhen as illustrated in Figure 1a. Assuming that there is no effect of the atmosphere and weather on the sun light intensity,

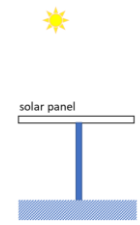
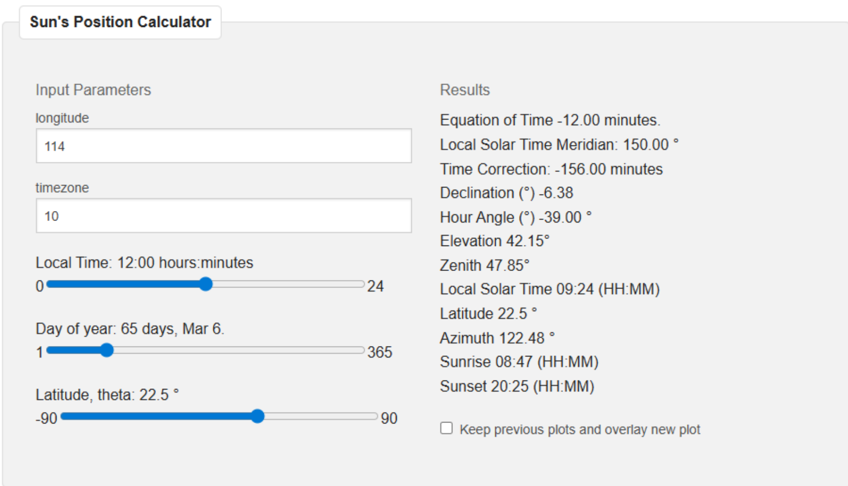
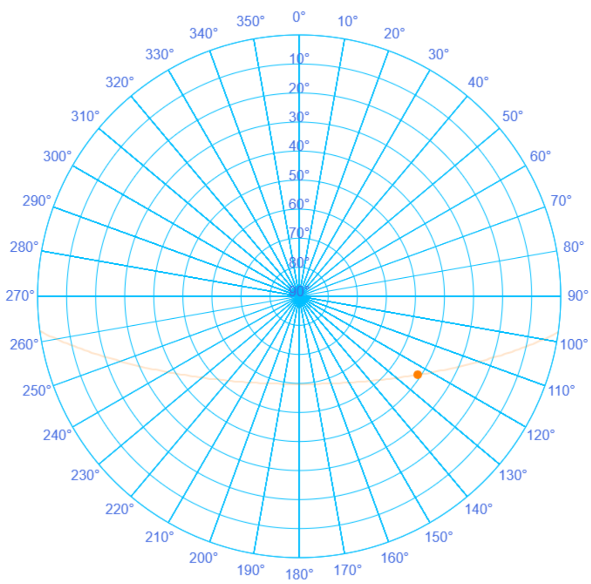


Figure 1a

(a) Plot the sun position throughout the day on the polar plot for today.

Shenzhen is located at 22.52°N, 114°E.





(b) On what day and month of the year and what time does the solar panel receive the highest power from the sun?

The solar panel receives the highest power from the sun on June 6th at 12:00 PM.

INPUTS

Location

Latitude	22.52	°
Longtiude	114	°

Date

Year	2026
Month	6
Day	6

Mean solar time

Hour	6
Minute	54
Second	10

Module installation

Tilt angle	0	°
Azimuth angle	0	°

OUTPUTS

During 2026:

Sunlight hours in year	4388.491

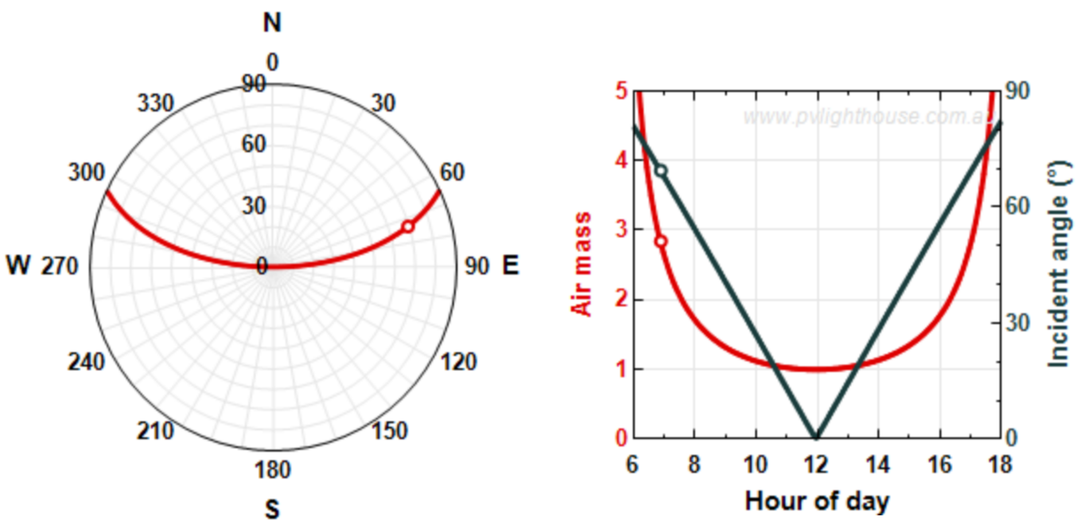
On 6-Jun-2026:

Sunlight hours in day	13:20:13
Sunrise (solar mean time)	5:18:48
Sunset (solar mean time)	18:39:01
Zenith angle at noon	1.213°
Air mass at noon	1.000

At 6:54:10, 6-Jun-2026:

Zenith angle	69.405°
Azimuth angle	73.077°
Air mass	2.843
Incident angle to module	69.405°

SOLAR PATH ON 6-Jun-2026



(c) If the solar panel is titled 30°from horizontal (Figure 1b) at that time, what is the relative change in the power received by the solar panel?

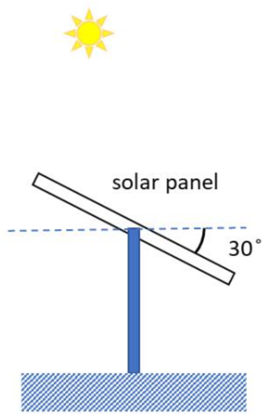


Figure 1b

INPUTS

Location

Latitude	22.52	°
Longtiude	114	°

Date

Year	2026
Month	6
Day	6

Mean solar time

Hour	6
Minute	54
Second	10

Module installation

Tilt angle	30	°
Azimuth angle	0	°

OUTPUTS

During 2026:

Sunlight hours in year	4388.491

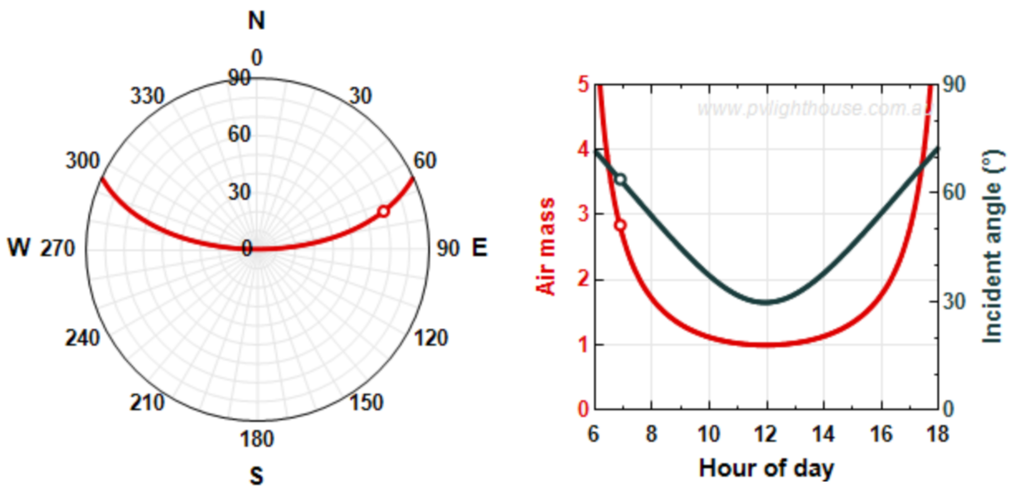
On 6-Jun-2026:

Sunlight hours in day	13:20:13
Sunrise (solar mean time)	5:18:48
Sunset (solar mean time)	18:39:01
Zenith angle at noon	1.213°
Air mass at noon	1.000

At 6:54:10, 6-Jun-2026:

Zenith angle	69.405°
Azimuth angle	73.077°
Air mass	2.843
Incident angle to module	63.840°

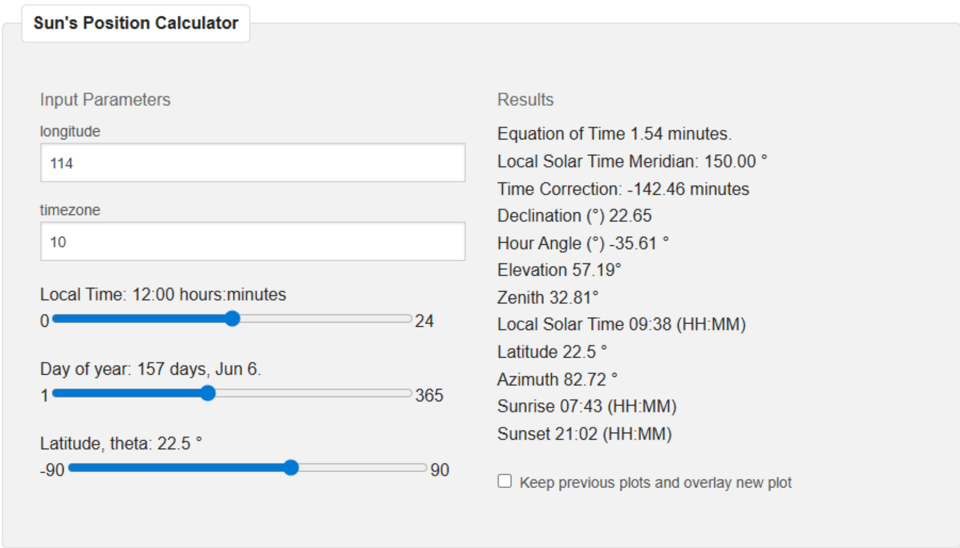
SOLAR PATH ON 6-Jun-2026

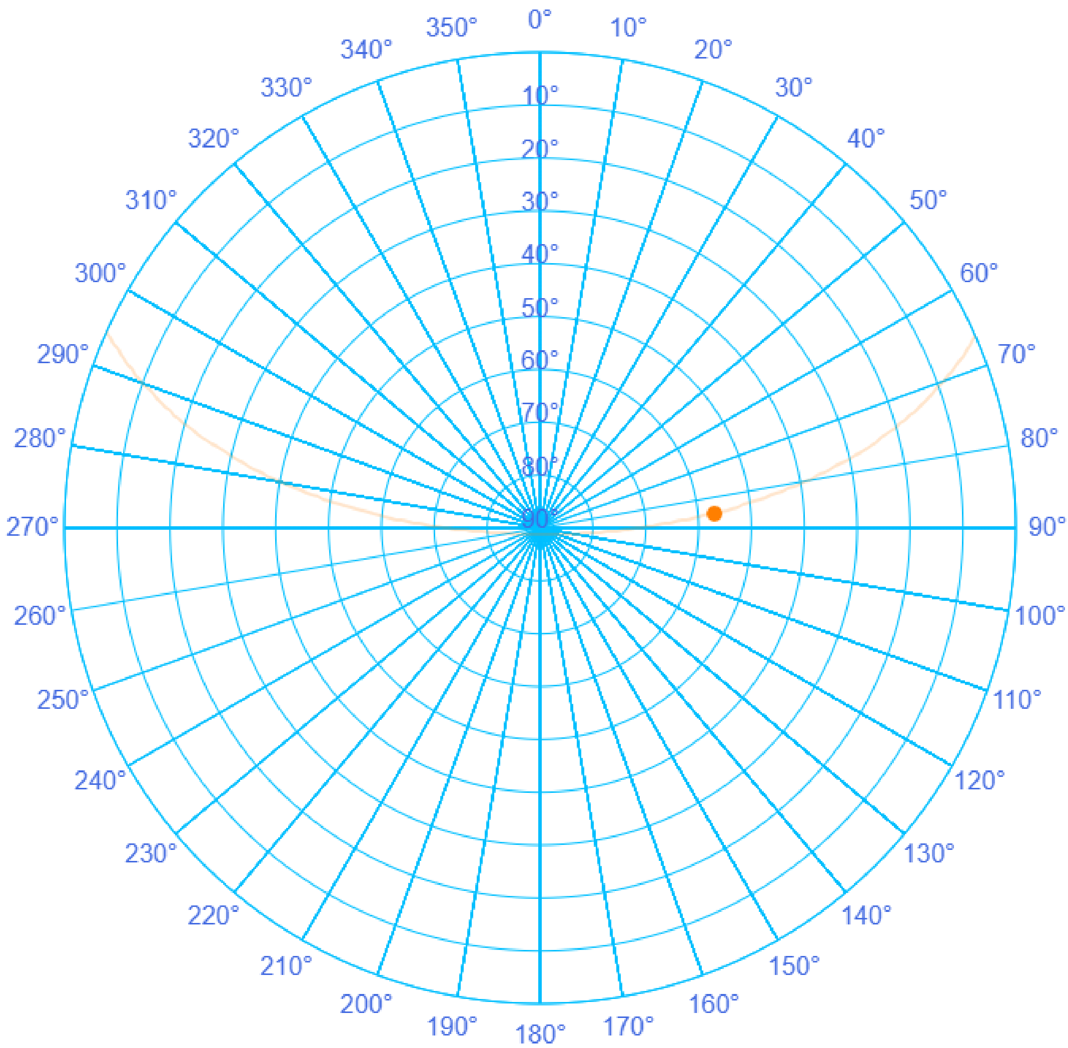


$$\text{Relative change} = \frac{cos(0°) - cos(30°)}{cos(0°)} = \frac{1-\frac{\sqrt{3}}{2}}{1} = 13.4\%$$

(d) Plot the sun's position throughout the day on the polar plot on that day.

As shown in the image, the yellow line represents the position of the sun.、





(e) How should position of the solar panel be adjusted (in terms of tilt angle and azimuth angle) at 9am, 12am, 3pm and 5pm of that day to receive the maximum power from the sun?

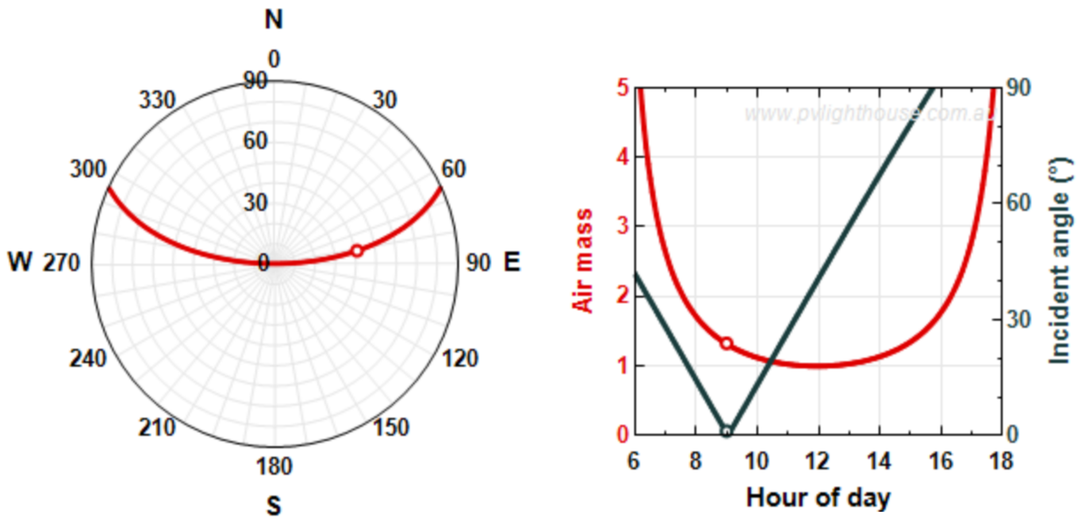
INPUTS

Location		
Latitude	22.52	°
Longtiude	114	°
Date		
Year	2026	
Month	6	
Day	6	
Mean solar time		
Hour	9	
Minute	0	
Second	0	
Module installation		
Tilt angle	40	°
Azimuth angle	80	°

OUTPUTS

During 2026:	
Sunlight hours in year	4388.491
On 6-Jun-2026:	
Sunlight hours in day	13:20:13
Sunrise (solar mean time)	5:18:48
Sunset (solar mean time)	18:39:01
Zenith angle at noon	1.213°
Air mass at noon	1.000
At 9:00:00, 6-Jun-2026:	
Zenith angle	41.103°
Azimuth angle	80.717°
Air mass	1.327
Incident angle to module	1.197°

SOLAR PATH ON 6-Jun-2026



9:00

Tilt Angle: 41.10°

Azimuth Angle: 80.717°

INPUTS

Location

Latitude	22.52	°
Longtiude	114	°

Date

Year	2026
Month	6
Day	6

Mean solar time

Hour	12
Minute	0
Second	0

Module installation

Tilt angle	0	°
Azimuth angle	0	°

OUTPUTS

During 2026:

Sunlight hours in year	4388.491

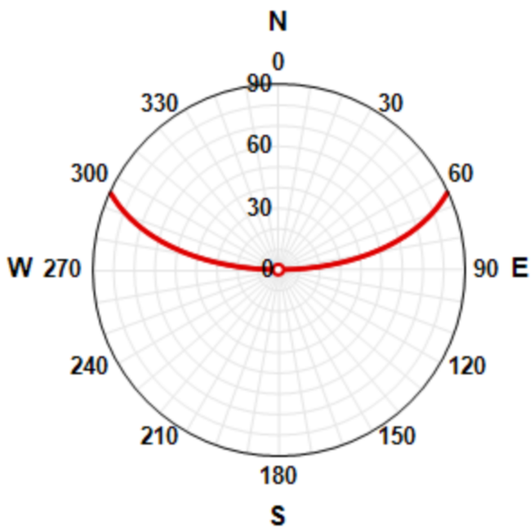
On 6-Jun-2026:

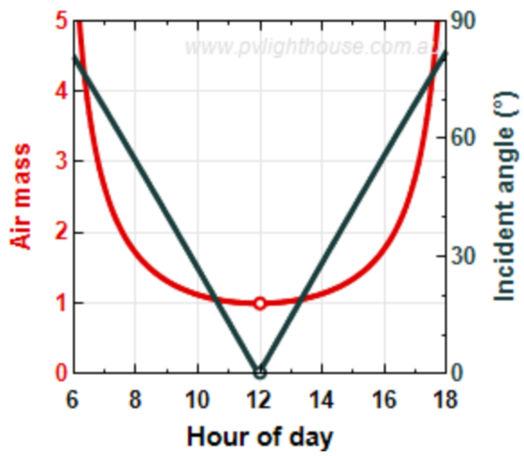
Sunlight hours in day	13:20:13
Sunrise (solar mean time)	5:18:48
Sunset (solar mean time)	18:39:01
Zenith angle at noon	1.213°
Air mass at noon	1.000

At 12:00:00, 6-Jun-2026:

Zenith angle	0.354°
Azimuth angle	311.363°
Air mass	1.000
Incident angle to module	0.354°

SOLAR PATH ON 6-Jun-2026





12:00

Tilt Angle: 0.354°

Azimuth Angle: 311.363°

INPUTS

Location

Latitude	22.52	°
Longtiude	114	°

Date

Year	2026
Month	6
Day	6

Mean solar time

Hour	15
Minute	0
Second	0

Module installation

Tilt angle	40	°
Azimuth angle	280	°

OUTPUTS

During 2026:

Sunlight hours in year	4388.491

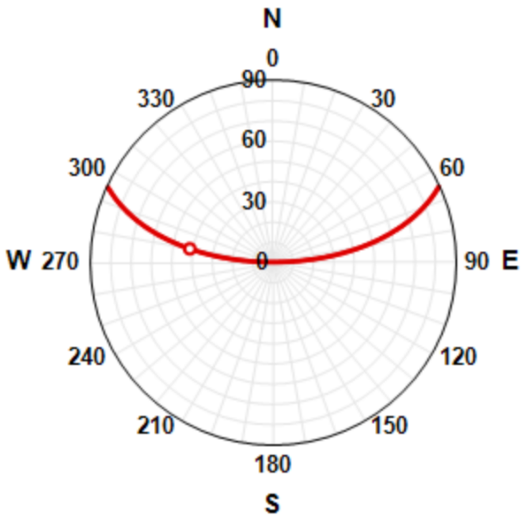
On 6-Jun-2026:

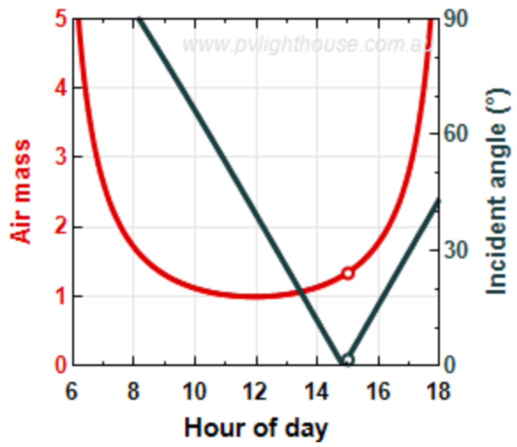
Sunlight hours in day	13:20:13
Sunrise (solar mean time)	5:18:48
Sunset (solar mean time)	18:39:01
Zenith angle at noon	1.213°
Air mass at noon	1.000

At 15:00:00, 6-Jun-2026:

Zenith angle	41.624°
Azimuth angle	279.441°
Air mass	1.338
Incident angle to module	1.665°

SOLAR PATH ON 6-Jun-2026





15:00

Tilt Angle: 41.624°

Azimuth Angle: 279.441°

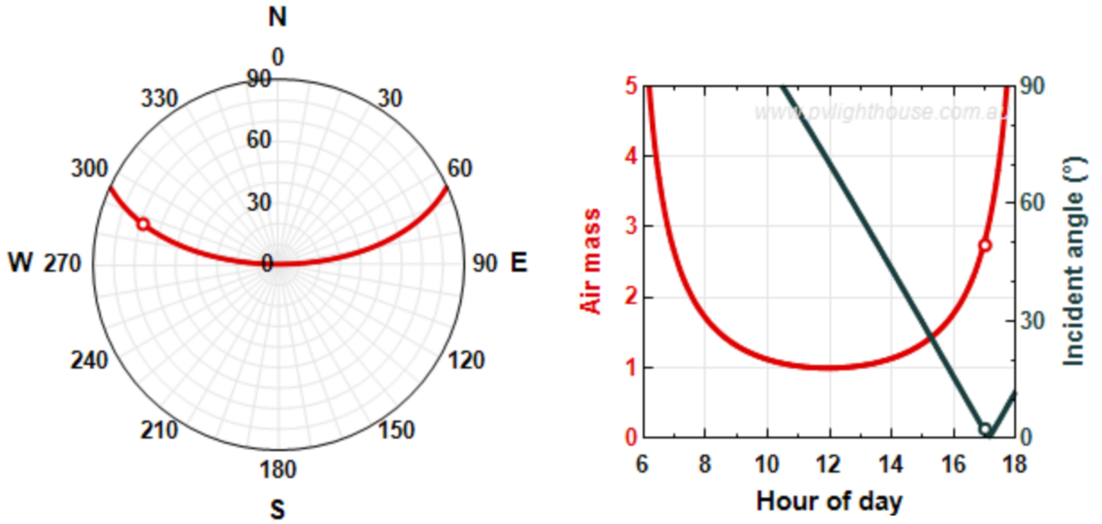
INPUTS

Location		
Latitude	22.52	°
Longtiude	114	°
Date		
Year	2026	
Month	6	
Day	6	
Mean solar time		
Hour	17	
Minute	0	
Second	0	
Module installation		
Tilt angle	71	°
Azimuth angle	287	°

OUTPUTS

During 2026:	
Sunlight hours in year	**4388.491**
On 6-Jun-2026:	
Sunlight hours in day	**13:20:13**
Sunrise (solar mean time)	**5:18:48**
Sunset (solar mean time)	**18:39:01**
Zenith angle at noon	**1.213°**
Air mass at noon	**1.000**
At 17:00:00, 6-Jun-2026:	
Zenith angle	**68.613°**
Azimuth angle	**286.717°**
Air mass	**2.742**
Incident angle to module	**2.402°**

SOLAR PATH ON 6-Jun-2026



17:00

Tilt Angle: 68.613°

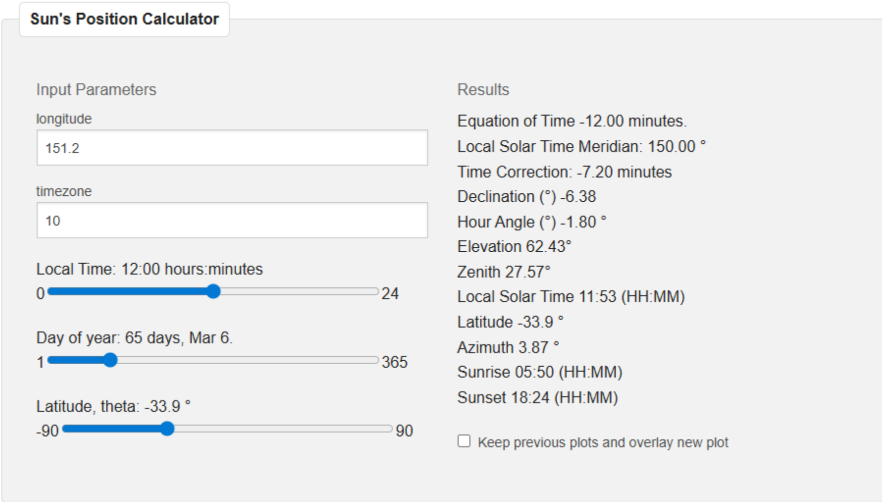
Azimuth Angle: 286.717°

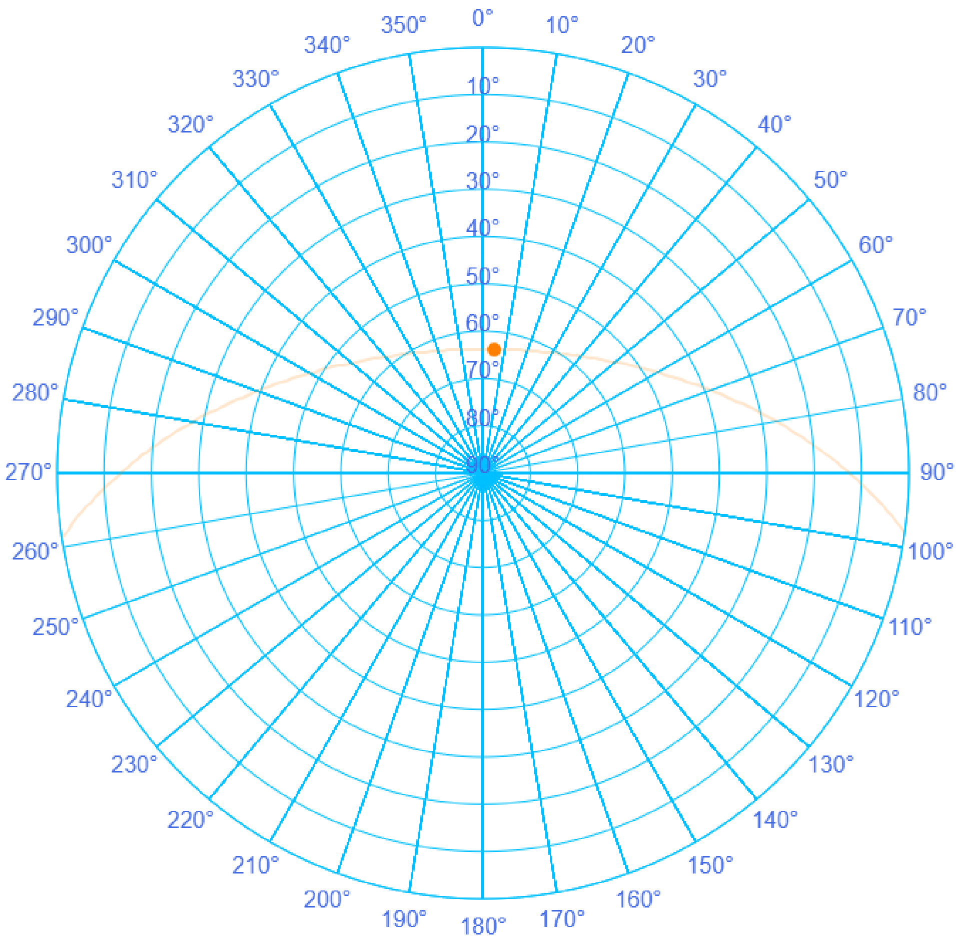
Quiz2

A solar panel is installed with tilt angle 30°(as in Figure 1b) at Sydney (Australia).

(a) Plot the sun position throughout the day on the polar plot for today.

Sydney is situated at 33.9°S, 151.2°E.





b） If the azimuth angle is set to 0°, plot the air mass and incident angle to the panel versus hour of the day.

Air mass and incident angle to the panel versus hour of the day.

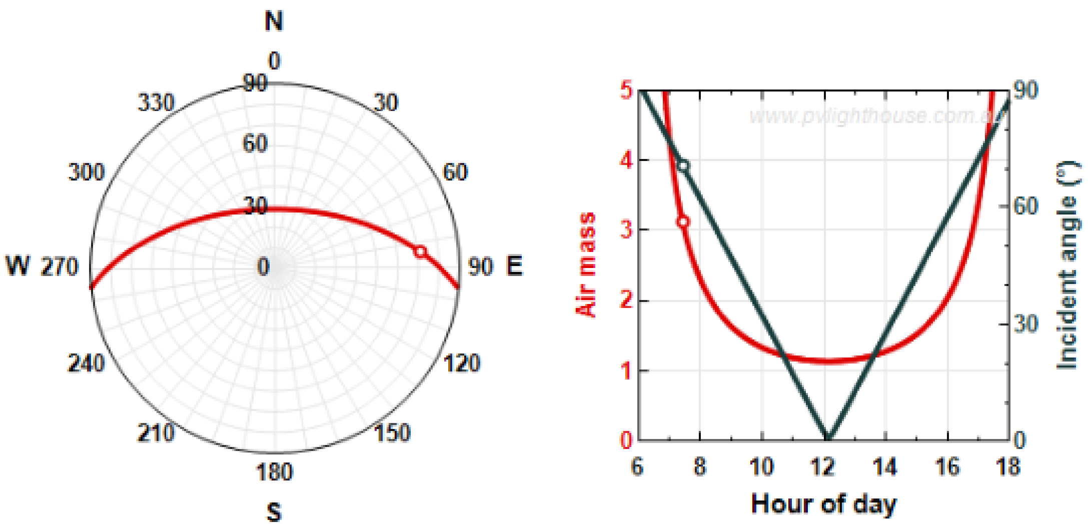
INPUTS

Location		
Latitude	-33.9	°
Longtiude	151.2	°
Date		
Year	2026	
Month	3	
Day	6	
Mean solar time		
Hour	7	
Minute	27	
Second	11	
Module installation		
Tilt angle	30	°
Azimuth angle	0	°

OUTPUTS

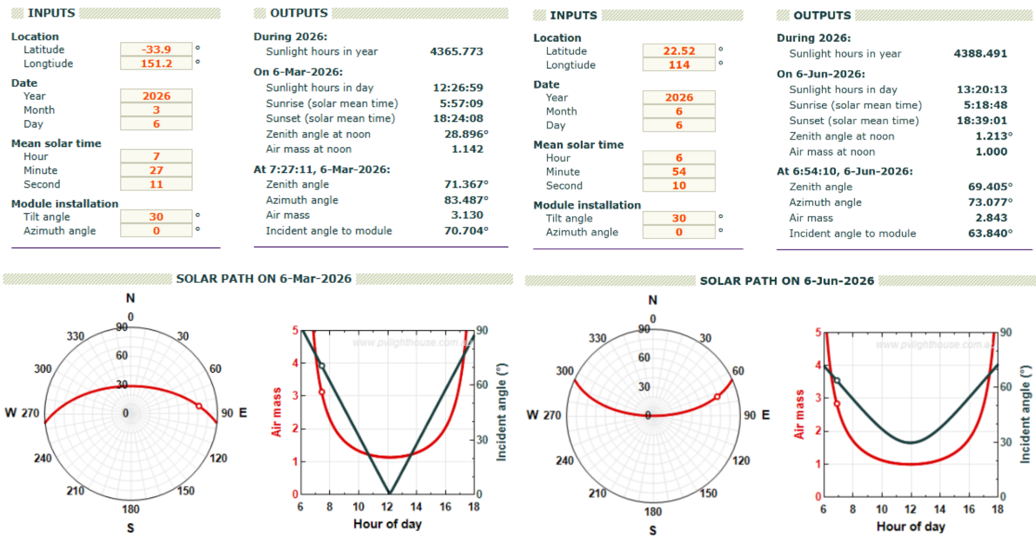
During 2026:	
Sunlight hours in year	4365.773
On 6-Mar-2026:	
Sunlight hours in day	12:26:59
Sunrise (solar mean time)	5:57:09
Sunset (solar mean time)	18:24:08
Zenith angle at noon	28.896°
Air mass at noon	1.142
At 7:27:11, 6-Mar-2026:	
Zenith angle	71.367°
Azimuth angle	83.487°
Air mass	3.130
Incident angle to module	70.704°

SOLAR PATH ON 6-Mar-2026



(c) Compare with the sun position plot that you observed in Shenzhen and comment on the observation.

Compare Sydney and Shenzhen



As depicted in the figure, with Sydney on the left and Shenzhen on the right, their respective locations in the Southern (33.9°S) and Northern (22.52°N) Hemispheres lead to completely different solar trajectories. This hemispheric contrast manifests as a clear north-south mirror symmetry in their solar polar diagrams.

(d) What should be the azimuth angle range of the solar panel for today?

Location		
Latitude	-33.9	°
Longtiude	151.2	°
Date		
Year	2026	
Month	3	
Day	6	
Mean solar time		
Hour	5	
Minute	57	
Second	9	
Module installation		
Tilt angle	30	°
Azimuth angle	0	°

During 2026:	
Sunlight hours in year	**4365.773**
On 6-Mar-2026:	
Sunlight hours in day	**12:26:59**
Sunrise (solar mean time)	**5:57:09**
Sunset (solar mean time)	**18:24:08**
Zenith angle at noon	**28.896°**
Air mass at noon	**1.142**
At 5:57:09, 6-Mar-2026:	
Zenith angle	**90.000°**
Azimuth angle	**96.185°**
Air mass	**n.a.**
Incident angle to module	**n.a.°**

Location		
Latitude	-33.9	°
Longtiude	151.2	°
Date		
Year	2026	
Month	3	
Day	6	
Mean solar time		
Hour	18	
Minute	24	
Second	8	
Module installation		
Tilt angle	30	°
Azimuth angle	0	°

During 2026:	
Sunlight hours in year	**4365.773**
On 6-Mar-2026:	
Sunlight hours in day	**12:26:59**
Sunrise (solar mean time)	**5:57:09**
Sunset (solar mean time)	**18:24:08**
Zenith angle at noon	**28.896°**
Air mass at noon	**1.142**
At 18:24:08, 6-Mar-2026:	
Zenith angle	**90.000°**
Azimuth angle	**264.059°**
Air mass	**n.a.**
Incident angle to module	**n.a.°**

The input is derived from the sunrise and sunset times given in the image, with the two extreme cases of azimuth angle being 96.185° and 264° respectively. The range of variation is 96.185°-360°-264°.

(e) What is the optimized tilt angle and azimuth angle of the solar panel to receive the highest power from sun at today noon?

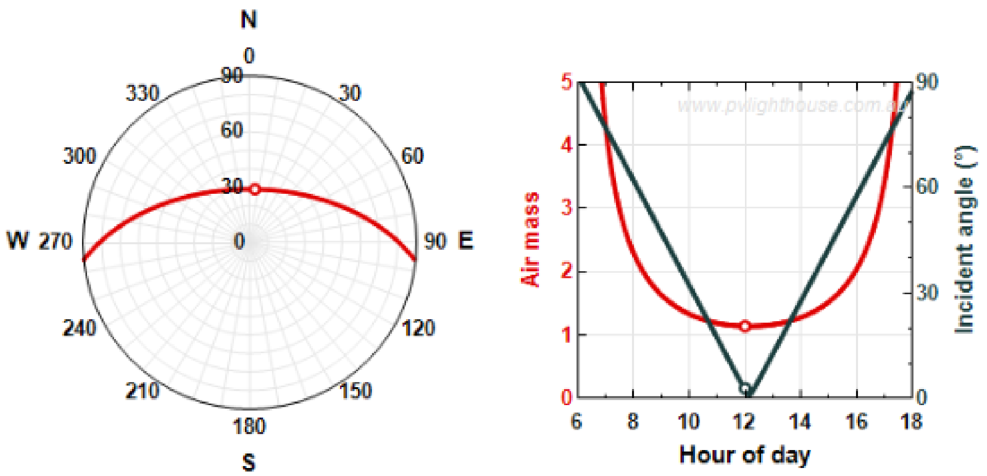
INPUTS

Location		
Latitude	-33.9	°
Longtiude	151.2	°
Date		
Year	2026	
Month	3	
Day	6	
Mean solar time		
Hour	12	
Minute	0	
Second	0	
Module installation		
Tilt angle	30	°
Azimuth angle	0	°

OUTPUTS

During 2026:	
Sunlight hours in year	4365.773
On 6-Mar-2026:	
Sunlight hours in day	12:26:59
Sunrise (solar mean time)	5:57:09
Sunset (solar mean time)	18:24:08
Zenith angle at noon	28.896°
Air mass at noon	1.142
At 12:00:00, 6-Mar-2026:	
Zenith angle	28.979°
Azimuth angle	5.620°
Air mass	1.143
Incident angle to module	2.948°

SOLAR PATH ON 6-Mar-2026



Tilt 28.979°

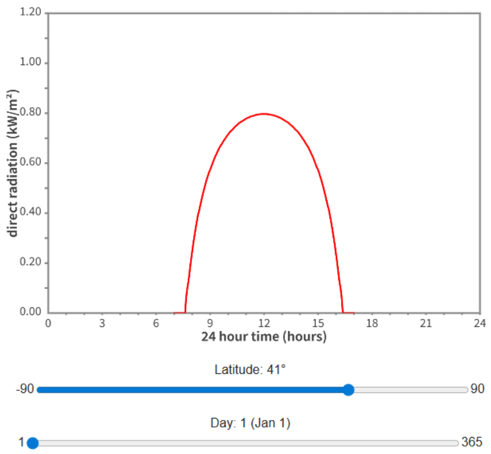
Azimuth 5.620°

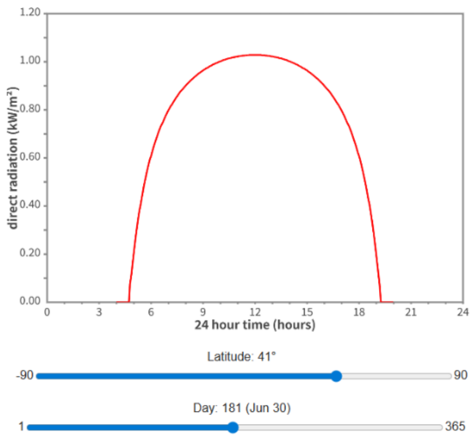
Quiz3

Two solar panels (A and B) are installed in New York City. The solar panel A is installed at tilt angle 0˚and the solar panel B is installed at tilt angle 65˚. Ignoring the local weather effects,

(a) Compare the solar insolation at New York City on 1st Jan and 30 June and calculate the solar intensity ratio between the two dates.

New York: 40.7°N, 74°W





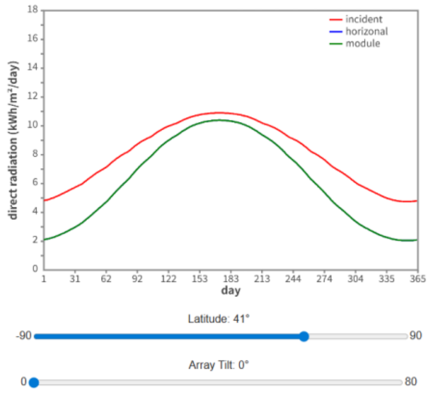
Intensity estimated at the highest point:

1st Jan: 0.797 kW/m^2

30th Jun: 1.029 kW/m^2

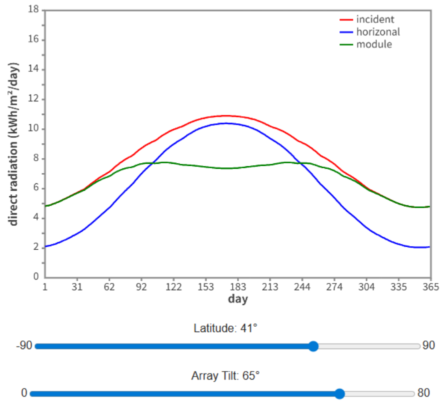
Ratio: 0.797/1.029 = 78%

(b) On which day(s) does the solar panel A receive the highest power from the sun?



The curve peaked around day 176, on **June 25th.**

(c) On which day(s) does the solar panel B receive the highest power from the sun?

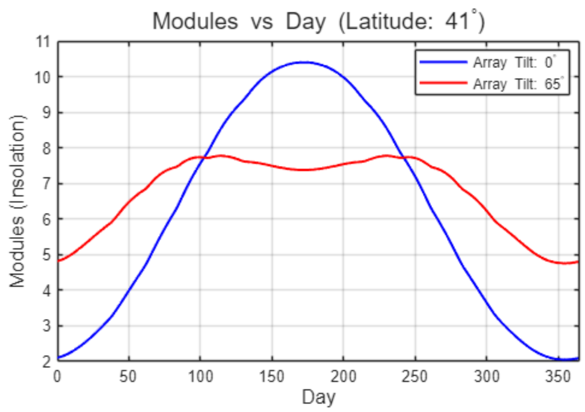


By observing the peak of the green curve and comparing it to the horizontal axis (Day), the two exact dates are approximately:

Around day 115, April 25th

Around day 230, August 16th.

(d) Plot the solar power received by each solar panel through out the year. In which period of the year, the solar panel A outperform the solar panel B? In which period of the year, the solar panel B outperform the solar panel A?



From the spring/autumn equinox to summer (about April to September):
The sun's path in the sky is relatively **high**. The **horizontally installed solar panel A** performs better than **panel B**. At this time, the **65° tilt angle** of panel B is too large, so the sunlight hits the surface at a very shallow angle, resulting in **low receiving efficiency**.

From the autumn/spring equinox to winter (about October to March):
The sun's path in the sky is **very low**. The **steep solar panel B** performs better than **panel A**. At this time, it can **better face the low-angle winter sunlight**, so it receives more solar energy.

(e) Should the tilt angle of the solar panel B be adjusted more than 65°?

No

As calculated in question (c), the **zenith angle** of the sun at its lowest point of the year (noon on the winter solstice) in New York City is about **64.15°**. A **tilt angle of 65°** already closely matches this extremely low solar angle. If the tilt angle is increased beyond **65°** (for example, **75° or even vertical at 90°**), the sunlight would strike the panel from slightly above even on the day when the sun is lowest in the year, which would reduce the received power. In addition, the performance loss would become even more significant during the rest of the year.

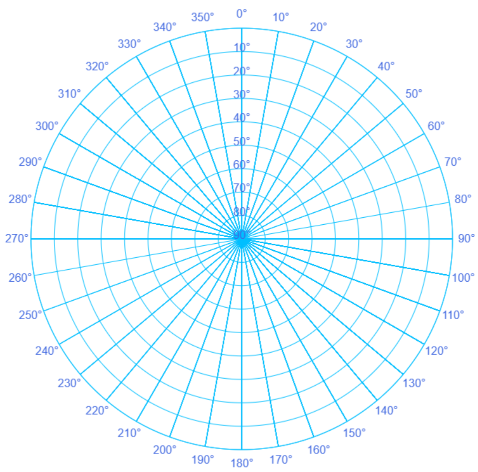
Quiz4

Suppose that you install a solar panel at the north pole and south pole.

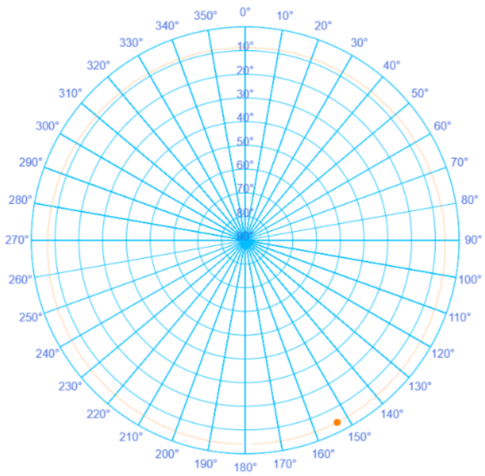
(a) Plot the sun position throughout the day on the polar plot for today.

North Pole, 90° North latitude; South Pole, -90° North latitude

North

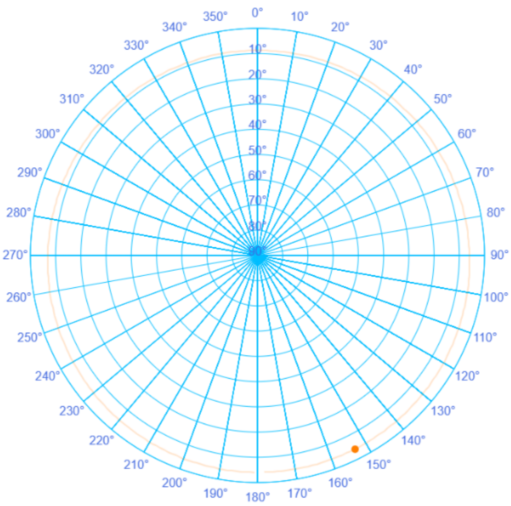


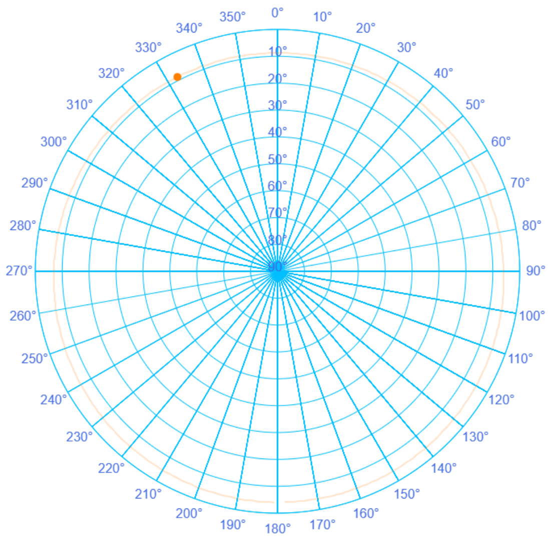
South



(b) Where is the sun position at mid night and noon?

At the poles, the sun's altitude is almost constant throughout the day and it orbits along the horizon, so the sun at noon and midnight is at the same altitude, but in completely opposite positions.





(c) Will you install the solar panel in horizontal position or vertical position?

Vertical installation.

At the pole, even in midsummer (the summer solstice), the **maximum solar elevation angle** is only about **23.45°**. This means that throughout the year, sunlight reaches the surface at a **very low angle near the horizon**. A **vertical installation** can better face these low-angle incoming rays and therefore **capture more solar energy**.

(d) At which tilt angle does the solar panel receive the lowest solar power?

The received power is **lowest when the panel is installed horizontally (tilt angle 0°).**

At the pole, the **solar elevation angle is very low**, so sunlight travels almost **parallel to the surface of a horizontally placed panel**. This very large **incident angle** leads to a **high reflection rate**, which greatly reduces the amount of **effective radiation absorbed by the solar panel**.

(e) How should the tilt angle of the solar panel be adjusted to receive the highest total solar power over the whole year?

To obtain the **highest total power throughout the year**, a simple **fixed tilt angle** is not sufficient.

Azimuth tracking: At the pole, the sun rotates **360° horizontally in the sky every day**. Therefore, the solar panel should be equipped with a **vertical-axis tracking system**, so that its front surface follows the sun and **rotates once in the horizontal direction every 24 hours**.

Seasonal tilt adjustment: The **tilt angle** of the panel should be adjusted with the **seasons** so that it always matches the **solar zenith angle**. At the **spring and autumn equinoxes**, when the sun just appears above the horizon (solar elevation angle **0°**), the panel should remain **fully vertical (90°)**. As the polar day progresses toward the **summer solstice**, when the solar elevation reaches its **maximum of 23.45°**, the tilt angle should **gradually decrease to about 66.55° (90° − 23.45°)**.